# Multilayer Network Analysis of Startup Cofounders and Investment Managers

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***Abstract*—Startup ecosystems are multilayer social networks where cofounders and investment managers connect through shared portfolio ties, education, work experience, and volunteering. This paper analyzes the 212 VC portfolio (54 companies) as a multilayer network to identify community structure and the institutional signals that explain it. A bipartite cofounder-investment manager network is built and three institutional layers (education, experience, volunteering) are attached using LinkedIn-derived data. The final graph includes 59 people (49 cofounders and 10 investment managers) and 93 multilayer edges. Using Louvain, greedy modularity, label propagation, and Girvan-Newman, stable communities appear in the 5-to-8 range with modularity up to about 0.49. Network statistics show a sparse but connected structure, with density 0.054, average degree 3.15, and diameter 6, plus strong disassortativity (r = -0.718) that signals a hub-and-spoke organization. Centrality rankings highlight investment managers as bridges across founder groups, while the cofounder side splits into sub-communities tied to shared institutional histories. Educational overlap concentrates around local universities with a smaller international tier, suggesting institutional pathways for access and trust formation. Implications for founders, investors, and policy are outlined, along with a reproducible workflow for multilayer network construction, community detection, and comparative analysis across layers.**

***Index Terms*—startup ecosystems, multilayer networks, venture capital, community detection, cofounder networks, investment managers, LinkedIn data**

## I. Introduction

## II. Related Work

## III. Data

### A. Data Sources

### B. Construction of the Dataset

## IV. Methodology

### A. Construction of the Networks

### B. Initial Analyses on the Networks

### C. Community Detection

## V. Results

## VI. Discussion

## VII. Limitations and Future Work

## VIII. Conclusion

## Acknowledgments

## References

## Appendices